

THE POWER OF VACCINATION

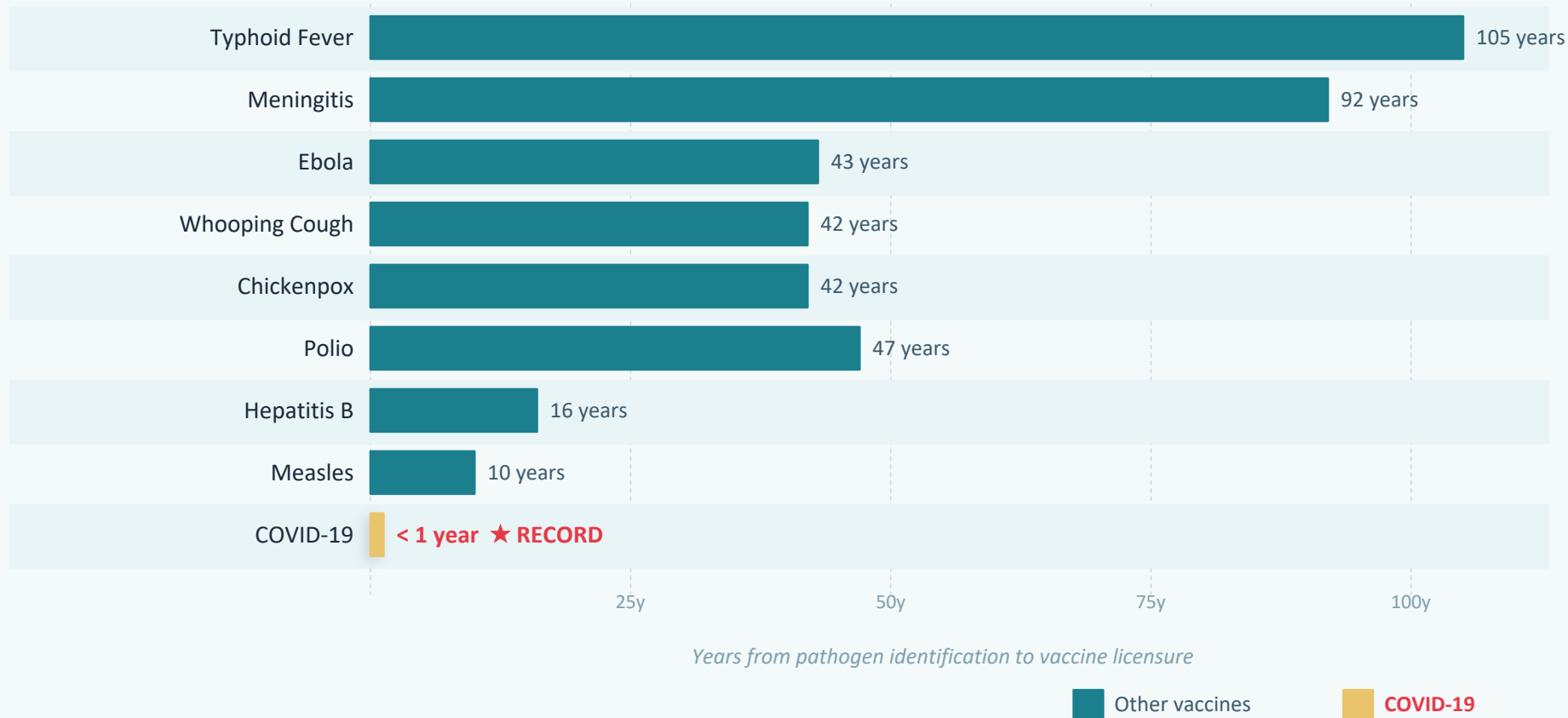
From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of vaccine impact on public health · 1796–2020

For public health professionals and policy analysts

Source: Our World in Data — ourworldindata.org/vaccination

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



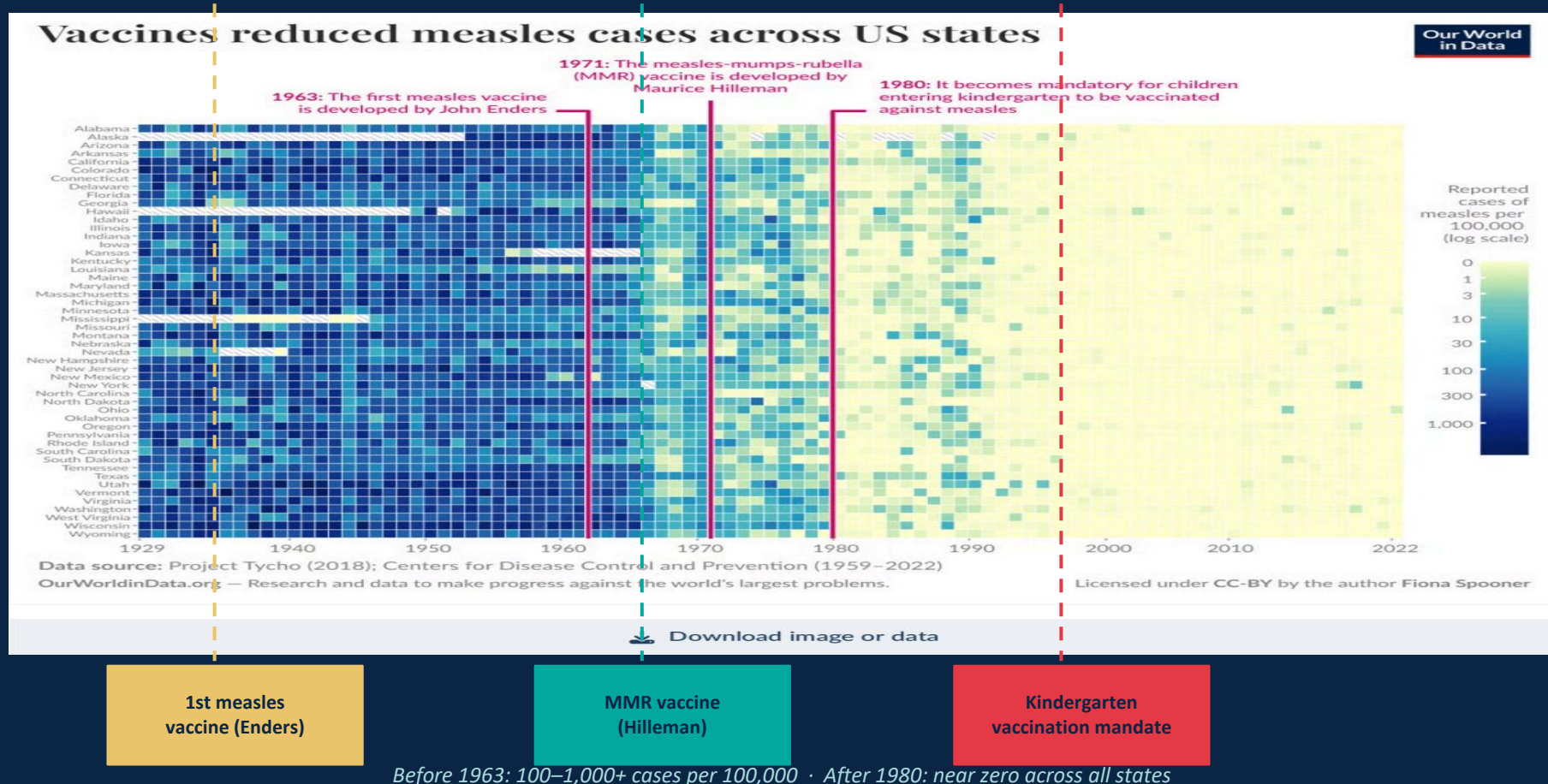
100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases (per million/yr)	Post-Vaccine Cases (per million/yr)	Case Reduction	Death Reduction
Diphtheria	158	0	100% ✓	100% ✓
Smallpox	250	0	100% ✓	100% ✓
Acute Polio	141	0	100% ✓	100% ✓
Measles	3,044	0.2	99.99%	100%
Rubella	242	0.04	99.98%	100%
Pertussis	1,534	52	96.6%	99.7%
Hepatitis A	465	51	89%	88.7%
Acute Hep. B	273	44	83.9%	83.6%
Pneumococcal	233	139	40.5%	31.3%

 100% eliminated

 Incomplete control — gaps remain

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



Smallpox: The Only Human Disease Eradicated Through Vaccination

1796

Edward Jenner creates first vaccine using cowpox to trigger immunity against smallpox.

1959

WHO passes resolution to eradicate smallpox globally.

1967

WHO launches intensified eradication campaign worldwide.

1977

Last natural smallpox case recorded in Somalia.

1980

WHO certifies global eradication — the only human disease eliminated through vaccination.

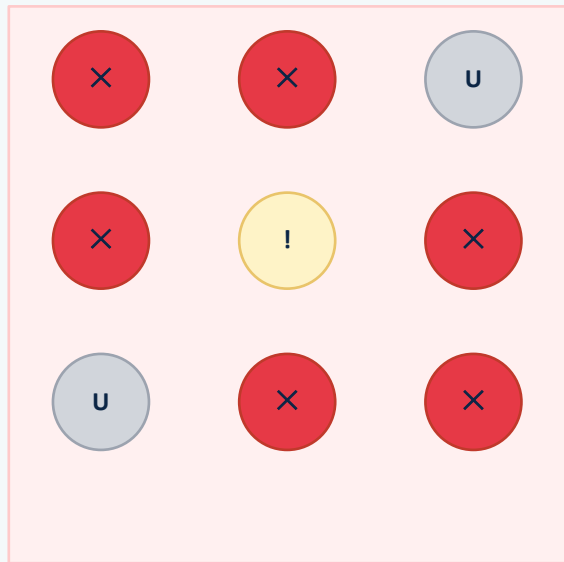
Why Smallpox Was Uniquely Eradicable

- Infected humans only — no animal reservoir to sustain transmission
- Visible, distinguishable symptoms made case identification straightforward
- Vaccine was cheap, stable, and highly effective
- Global political will aligned behind WHO campaign

Certified eradicated 1980 · No human cases since 1977

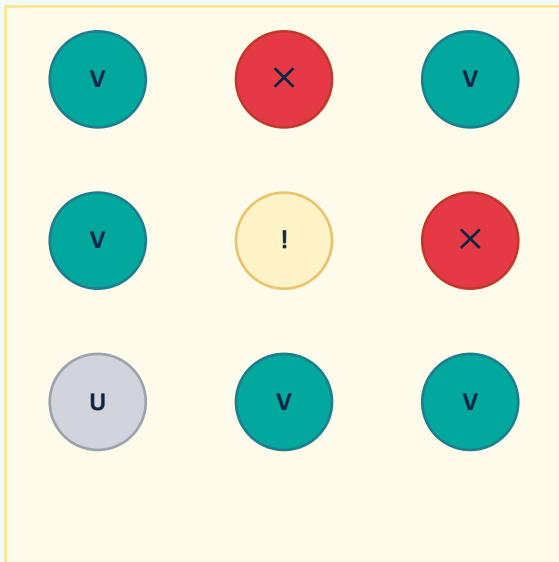
Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

No Vaccination



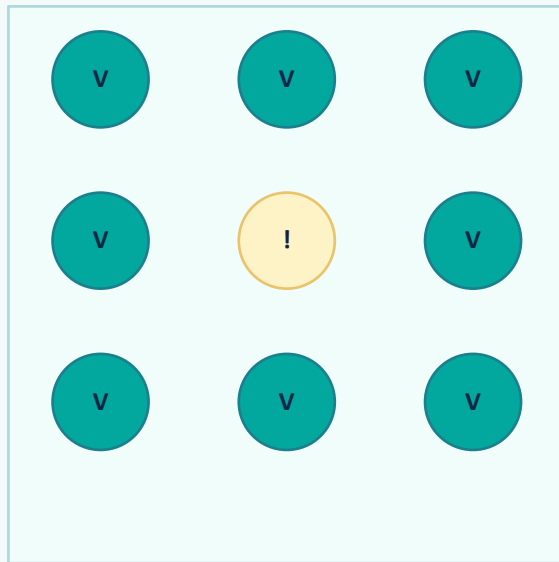
Disease spreads freely

Partial Vaccination



Sporadic spread — some risk remains

Herd Immunity Achieved



Pathogen cannot spread — vulnerable shielded

 Vaccinated

 Unvaccinated

 Vulnerable (cannot be vaccinated)

 Infected

Vulnerable individuals: newborns, immunocompromised patients, cancer patients on chemotherapy

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal Disease

Case reduction	40.5%
Death reduction	31.3%

Largest remaining gap

Acute Hepatitis B

Case reduction	83.9%
Death reduction	83.6%

Still ~44 cases/million/yr

Hepatitis A

Case reduction	89%
Death reduction	88.7%

Still ~51 cases/million/yr

◆ Access & Supply

Not everyone globally has access to vaccines; supply shortages put children at risk.

◆ Scientific Literacy

Poor understanding of vaccines can reduce trust and uptake in communities.

◆ Coverage Gaps

Incomplete vaccination coverage leaves populations vulnerable to preventable disease.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

01

100% Elimination Achieved

Diphtheria, smallpox, and acute polio have been 100% eliminated in the US — cases and deaths reached zero.

02

Development Timelines Collapsed

From 105 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.

03

Herd Immunity Protects Communities

High vaccination rates shield those who cannot be vaccinated: newborns, immunocompromised patients, and cancer patients.

04

Smallpox: The Only Human Disease Eradicated

Certified eradicated in 1980 after a global WHO campaign — no natural cases since 1977.

05

Coverage Gaps Persist

Pneumococcal (40.5% case reduction), hepatitis A (89%), and hepatitis B (83.9%) still require stronger coverage.

Call to Action

- Strengthen global vaccine supply chains to prevent shortages
- Expand coverage for pneumococcal, hepatitis A & B to close remaining gaps
- Invest in public trust through science communication and community engagement
- Leverage mRNA platform innovation to accelerate future outbreak response